

Neuromatrix theory of pain

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1 Pain, nociception, neurosignatures, and the body–self neuromatrix

Pain is generally subjectively perceived as a direct, intense, and unitary experience. However, despite this apparent simplicity, a comprehensive scientific explanation of the central nervous system (CNS) processes that generate the subjective experience of pain still remains elusive. Indeed, although two centuries of systematic neurophysiological studies have succeeded in mapping out many of the pathways and relays through which information about potentially harmful stimuli are conveyed from the periphery to the brain (Perl 2007), we still lack a clear understanding of the cerebral processes that convert these raw sensory signals into the subjective experience we feel as pain. Indeed, pain refers to the subjective “emotional and sensory experience associated with actual or potential tissue damage, or described in terms of such damage” (IASP 2014; see also Chapter 31, this volume). Therefore, by definition, pain implies consciousness. (Cf. Chapter 17, this volume.) By contrast, the term “nociception” is reserved to designate the “neural processes of encoding noxious stimuli” (IASP 2014), which may not be necessarily associated with either consciousness or pain. From that perspective, the question of how the brain generates the subjective experience of pain can be seen as the quintessential mind–body problem of how subjective experience arises from matter.

One simplistic way to solve the problem would be to simply deem it out of neuroscience’s reach: Neuroscience can explain nociception, but not pain. Indeed, pain as a subjective experience cannot be ontologically reduced to neuronal activity: there isn’t any pattern of neuronal activity that could possibly *be* pain (Searle 2007). Still, it is reasonable to think that pain is caused by the CNS. In other words, although pain cannot be *ontologically* reduced to any one

state of neuronal activity, it may be *causally* reduced to neuronal activity. One possibility could be that all pain is caused by nociception. However, an emerging consensus is that nociception doesn't have a 1 : 1 relationship with pain, and that it is thus possible to have pain without nociception, and nociception without pain. Indeed, pain behaviors in animals and pain reports in humans can be influenced by brain processes that do not appear to be related to changes in nociception (Johansen and Fields 2004; Navratilova and Porreca 2014). Thus, there is likely another layer of neuroscientific phenomena between pain and nociception that may explain how brain activity causes pain, above and beyond (and perhaps in the absence of) nociception.

The kinds of neural processes that comprise this “intermediate layer” were first imagined by Ronald Melzack almost twenty years ago (Melzack 1999). Indeed, Melzack was puzzled by the fact that the pain experienced by patients with phantom limbs presented all the qualities of “normal” pain, despite the obvious absence of peripheral and/or spinal nociceptive activity (Baron and Maier 1995; Birbaumer et al. 1997). He concluded that the pain therefore had to be caused by the same cerebral processes that are responsible for normal pain perception. Melzack proposed that the experience of pain was generated by the flow of activity within a widespread network of convergent/divergent loops between the thalamus, cortex, and limbic system. He thought that the general function of this network, which he called the “body–self neuromatrix,” was to produce a unified representation of the body and of the self. Moreover, he suggested that the various bodily states generated by the neuromatrix each had a particular “neurosignature,” i.e., a characteristic pattern of neuronal activity stemming from the interaction between environmental inputs and the pre-existing state of the neuromatrix. According to Melzack, there is therefore no “pain center” in the brain, but rather a general-purpose body–self neuromatrix capable of representing a wide array of bodily states, of which pain is just one particular exemplar. Whereas Melzack's terminology had a profound impact on subsequent brain-imaging studies of pain, his ideas have been largely misinterpreted and the core hypothesis of his neuromatrix theory of pain – that pain is generated by a specific pattern of activity within a general-purpose system – still remains to be tested using modern neuroimaging techniques.

2 From neuron to voxels: a brief history of brain imaging of pain

Before the advent of functional brain imaging in the 1980s, physiologists had already mapped out several ascending pathways conveying nociceptive signals from the dorsal horn of the spinal cord up to the brainstem, thalamus, and a

number of cortical and sub-cortical sites. The presence of neurons with nociceptive properties (sensitivity to potentially tissue-damaging high-intensity stimuli) was also beginning to be reported in a number of cortical and subcortical structures including the hippocampus, thalamus, amygdala, anterior cingulate, insula (Ins), and primary somatosensory cortex (Kenshalo et al. 2000; Millan 1999; Vogt and Sikes 2000). However, the techniques at their disposal generally only allowed examination of one local recording site in one brain region at a time. Typically, experiments would target a brain region and systematically test the response profiles of hundreds of neurons, isolated individually, in search of neurons that gradually increased their firing rate as a function of stimulus intensity (wide dynamic range neurons), or that only responded to high-intensity stimulation (nociceptive-specific neurons; Kenshalo et al. 2000). This approach became increasingly cumbersome as the targeted areas became larger and contained fewer and more dispersed nociceptive neurons (e.g., 9/125 tested neurons in the anterior cingulate cortex, ACC; Hutchison et al. 1999). Additionally, it remained unclear how nociceptive stimulus encoding in isolated cells relates to pain experience. In other domains, a growing set of studies focusing on population coding was suggesting that perceptual and motor processes are an emergent property of the overall pattern of activity in populations of neurons, rather than activity in any single neuron (Georgopoulos et al. 1986). Thus, it was (and is) uncertain how activity in nociceptive neurons is related to subjective pain perception.

With the advent of functional brain imaging in the late 1980s, researchers began to have the tools to image how the entire brain responded to noxious stimuli. The first technique to be used was positron emission tomography (PET), which was soon replaced by functional magnetic resonance imaging (fMRI). Both techniques use indirect measures of neuronal activity – cerebral blood flow for PET and blood oxygenation for fMRI – to measure brain activity in small parcels of the brain called voxels. In a typical fMRI experiment, each voxel has a volume of approximately 8–64 cubic millimeters. A typical 3-millimeter $\times$ 3-millimeter $\times$ 3-millimeter voxel (27 cubic millimeters) contains on the order of 5.5 million neurons (Logothetis 2008). Using these techniques, several studies found that high-intensity noxious stimuli (thermal, electrical, mechanical, or chemical) systematically produced more activity than low-intensity innocuous stimuli in a set of regions that included the thalamus, ACC, Ins, primary and secondary somatosensory cortices (S1 and S2) and prefrontal cortices (PFC) (Apkarian et al. 2005). Inspired by Melzack's methodology, researchers soon started to refer to this collection of regions as the "pain matrix" (Ingvar 1999; Jones 1998).

While many researchers mainly employed the term as shorthand for the regions generally activated by pain, the concept of the pain matrix quickly began to implicitly designate the brain's "pain center" (Ingvar 1999). More specifically, it was thought that each part of the pain matrix mapped onto different aspects or qualities of pain (e.g.: localization, intensity, unpleasantness, motivation, etc.), and that their combined activation is what creates the full experience of pain. In other terms, it was thought that it was the global pattern of activation across all of the pain matrix's regions that created pain, and not activity in any of its subregions taken independently. Based on prior neurophysiological evidence suggesting a parsing of the spinothalamic tract into a lateral/sensory system and a medial/emotional system, it was suggested that S1 and S2 subserved the sensory dimension of pain, while the Ins and ACC were responsible for pain's unpleasantness (Price 2000). Finally, the prefrontal cortex was thought to be involved in "secondary" cognitive and emotional reactions to pain, such as rumination, anxiety, and anger, which depend on higher-order appraisals of the meaning of the pain for the individual (Price 2000, [Chapter 9](#), this volume). While this general architecture agrees relatively well with neurophysiological evidence (Millan 1999) and lesion data (Danziger 2006; Ploner et al. 1999), there is no evidence that any of the regions, or combination of regions, of the pain matrix is specific to pain. Indeed, several brain-imaging studies have shown that most, if not all, structures of the pain matrix are sensitive to different kinds of salient events that are not necessarily painful (Yarkoni et al. 2011; [Chapter 6](#), this volume).

3 From voxels to pain: reverse inference and the problem of pain specificity

Debates around pain's specificity are as old as research on pain physiology itself (Perl 2007). Unlike the other senses that process a specific type of physical stimulus, like vision and light or audition and air waves, pain can be generated by a variety of somatosensory inputs (thermal, mechanical, chemical). It was therefore proposed that pain may be simply caused by the vigorous activation of nervous pathways normally concerned with somato- and viscerosensation. However, based on observations that hemisections of the spinal cord produced a double-dissociation between pain and tactile sensation, it was also argued that pain may very well be a specific sense distinct from normal somatosensation. Up to this day, this opposition between specificity and intensity theories of pain still fuels debates regarding how the CNS generates pain. Needless to say, the question of pain specificity also drove important controversies regarding the

interpretation of brain-imaging data. At the core of these controversies lies the problem of reverse inference, i.e., the conclusion that a certain experience (e.g., pain) has been elicited based on the presence of activity in regions shown to be activated by that same experience (e.g., pain-matrix regions) in previous studies, even if these regions may also be involved in a variety of other experiences or processes (e.g., attention, negative affect, conflict, etc.).

One example of reverse inference is the use of activity in the pain matrix to infer effects of non-pharmacological interventions on “pain processing” (Bushnell et al. 2013). Typically, brain modulation by conditions including hypnosis (Rainville et al. 1997), placebo/expectations (Wager et al. 2004), emotion (Roy et al. 2009), distraction (Tracey et al. 2002), and others, was used to corroborate effects on pain reports and infer meaningful changes in pain mechanisms. Moreover, some of these studies also found increased activity in brain neurochemical systems (Wager et al. 2007) and regions involved in descending pain modulatory systems, such as the rAAC (rostral ACC) or PAG (periaqueductal gray) (Tracey et al. 2002; Wager et al. 2004), which was interpreted as evidence that these interventions may alter spinal nociceptive processes. While it is perfectly plausible that non-pharmacological interventions influence activity in the same pain-processing regions that respond to noxious inputs, these results also tended to be interpreted by the greater scientific community as a proof that these interventions did not just induce biases in pain reports, but were really having an effect on subjective pain perception. Unfortunately, for decreased activity in pain-matrix regions to provide evidence that these interventions were really effective in decreasing pain requires that pain-matrix activity be specific to pain-processing, an interpretation that is not supported by analyses of imaging data across many types of tasks (Yarkoni et al. 2011).

Another example of reverse inference without evidence for specificity came from studies in social neuroscience examining the cerebral substrates of social rejection and pain empathy. When these studies were published, the concept of the pain matrix as a specific set of pain-processing regions had become insidiously entrenched in the minds of the greater cognitive neuroscience community. Therefore, it seemed natural to conclude that activity in pain-matrix regions (ACC, Ins, etc.) could be interpreted as proof that “pain systems” were involved when individuals experience social rejection (Eisenberger and Lieberman 2004) or observe others in pain (Singer et al. 2004). Again, the problem with these otherwise plausible conclusions is that other explanations, which do not rely on the shared involvement of pain systems, are equally valid. Indeed, because each of the structures of the pain matrix are involved in many

processes other than pain, it is impossible to conclusively infer the presence of pain from activity in any of these regions (Poldrack 2006). In a strong critique of the pain matrix's alleged specificity, Iannetti and Mouraux (2010) proposed that regions of the pain matrix could be simply responsive to salient events in general. Indeed, they noted that intense or unexpected visual and auditory stimuli also strongly activated the thalamus, ACC, and Ins. Moreover, they found that in addition to activity in general salience-processing regions, intense non-painful tactile stimuli also activated S1, rendering their global pattern activity undistinguishable from the pattern associated with painful stimuli. On the basis of these findings, Iannetti and Mouraux concluded that activity in pain-matrix regions is not specific to pain, and could be better accounted for by a more general salience-processing interpretation.

However, this leaves intact the question of the cerebral substrates of pain: how can the experience of pain, which we feel as distinct from other salient events (e.g., receiving an unexpected reward), be caused by non-specific activity in a general salience-processing system? Our view is that it cannot. As originally suggested by Melzack, one possibility could be that the pain matrix constitutes a general-purpose architecture (i.e., the body-self neuromatrix) whose billions of neurons represent many states – and that pain constitutes a specific state (i.e., a neurosignature) or collection of states of the system. Indeed, nociceptive projections to the brain appear to converge in relatively well-defined subparts of pain-matrix regions, making the existence of such distinctive pain neurosignatures quite likely. For example, spinothalamic tract (STT) projections have been shown to target specific subparts of the posterior granular insular cortex (Ig; 41 percent of all projections), parietal operculum (S2; 29 percent of all projections), and cingulate cortex (24 percent of all projections) (Dum et al. 2009).

Interestingly, the relatively small insular (Ig) and parieto-opercular (S2) regions that receive most of STT projections appear to be relatively specific to nociceptive input. Indeed, they have been shown to respond to noxious heat, but not to innocuous warmth, cold, brushing, or proprioceptive inputs (Mazzola et al. 2012a). Moreover, they appear to be critical for the sensory-discriminative aspects of pain: their lesion results in a selective contralateral loss of temperature and pain sensation (Garcia-Larrea et al. 2010). Finally, these are the only two known cortical regions capable of generating painful sensations when stimulated (Mazzola et al. 2012b), suggesting that they may very well stand at the interface of nociceptive processing and pain perception. Still, neither of these two structures seems to be necessary or sufficient for pain. Indeed, a large proportion (more than 60 percent) of patients with operculo-insular lesions will develop

central pain in proportion with their thermo-algesic deficits (Garcia-Larrea et al. 2010), suggesting that these patients' pain must be generated by neuronal activity outside of these two operculo-insular regions. Moreover, nociceptive sensations, such as "burning," "pricking," or "throbbing" sensations are not painful per se, because they are not necessarily accompanied by the negative affect that characterizes pain; it is only when they are interpreted as potentially tissue damaging and associated with negative affect that the overall experience can be labeled as "pain" (IASP 2014). Consequently, lesion of the cingulate and prefrontal structures responsible for pain unpleasantness, or of the white matter tracts connecting these regions to the operculo-insular region, can cause an intriguing dissociation between pain sensation and affect whereby nociceptive sensations are no longer perceived as unpleasant. Interestingly, this condition has been called "pain asymbolia" by the neurologists who first described it to signify that pain had lost its meaning (Danziger 2006). Thus, although the operculo-insular region appears to be very important for the processing of ascending nociceptive inputs, it does not have a 1:1 relationship with the subjective experience of pain.

4 From voxels to signatures: identifying the cerebral representation of pain

Unlike other sensations, pain always implies some evaluation of the meaning of the raw nociceptive sensation in order for the experience to have all the properties of pain, i.e., an emotional and sensory experience associated with actual or potential tissue damage. Consequently, pain cannot be simply generated by the passive reception of ascending nociceptive inputs to the brain; it has to emerge from the concerted activity of large-scale networks of neurons spanning sensory, emotional, and cognitive-evaluative regions of the brain (Melzack 1999; Tononi and Koch 2015). Moreover, as originally proposed by Melzack (1999), it is highly unlikely that any of the processes binding together the different aspects of pain are per se pain-specific. Indeed, it could be argued that most, if not all, of our conscious subjective experiences are the result of combined sensory, emotional, and cognitive processes (Tononi and Koch 2015). It is therefore not surprising that previous attempts to link pain to a specific macroscopic network of pain-processing structures have failed (Iannetti and Mouraux 2010).

However, as mentioned previously, it remains possible that fine-grained patterns of activity within a more general-purpose system may be relatively specific to pain. Unfortunately, the traditional fMRI analyses used in prior brain-imaging studies of pain typically examine each of the brain's voxels

independently one from another, thereby ignoring the important information that may potentially lie in distributed patterns of neuronal activity and connections among regions. It is only with the recent development of multivariate pattern analyses (MVPA) that researchers were endowed with analytic tools capable of assessing the predictive power of distributed patterns of fMRI activity (Haxby et al. 2014). In a typical MVPA analysis, machine-learning algorithms are used to train multivariate classifiers – patterns of predictive weights distributed over hundreds to thousands of voxels – that aim to predict a certain outcome of interest (e.g., distinguish the presence of a face vs. an object). The predictive accuracy of these trained classifiers is then tested in a separate holdout validation set using cross-validation procedures. The resulting classifiers are therefore optimally tuned for making predictions for new, unseen data points. Using this principle, a wealth of applications are now being developed that allow researchers to “mind-read” or “decode” perceptions, intentions, and actions from patterns of brain activity (Haxby et al. 2014).

Inspired by these studies, we recently developed and tested a pain-predictive pattern that we have called the “neurologic pain signature” (NPS) (Wager et al. 2013). The NPS was originally trained to predict the subjective pain ratings associated with four levels of thermal stimulation, from innocuous warmth to highly painful heat. It is therefore designed to track the pain that arises from changes in temperature. As expected, NPS predictions were extremely accurate, with an average prediction error of 0.96 (plus or minus 0.33) on a 9-point visual analogue scale. Unsurprisingly, the NPS pattern of pain-predictive weights comprised positive weights in regions of the pain matrix such as the ACC, Ins, S2, and thalamus, and negative weights in default mode structures frequently deactivated by pain, such as the ventromedial prefrontal cortex (vmPFC) and precuneus (see [Figure 7.1](#)). However, contrary to earlier conceptions of the pain matrix, it wasn’t the overall macroscopic pattern of regions of the signature that predicted pain, but rather the fine-grained pattern of weights within each structure.

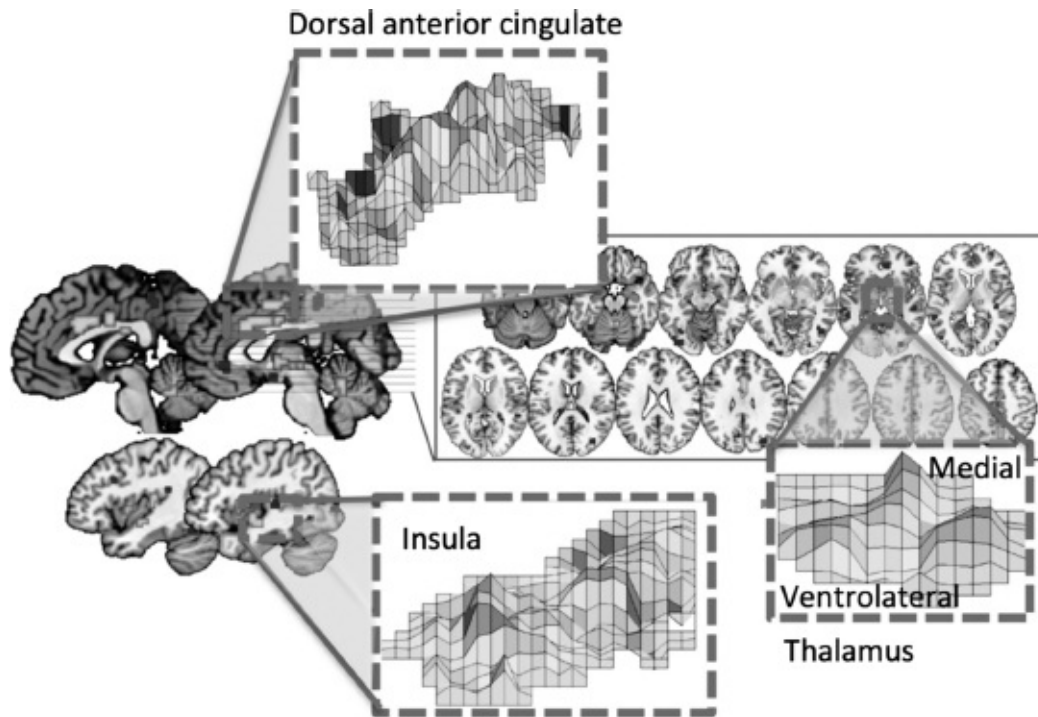


Figure 7.1 The neurologic pain signature map, consisting of voxels in which activity reliably predicted pain. Warm/cold colors (see online, www.routledge.com/9781138823181) indicate positive/negative weights.

Source: Zaki et al. (2014).

We then examined how the NPS behaved across various independent data sets. We began by testing the NPS sensitivity to six predetermined levels of thermal stimuli, ranging from innocuous warmth to painful heat (44.3, 45.3, 46.3, 47.3, 48.3, and 49.3° C; stimuli were administered on the left forearm with a 3-centimeter × 3-centimeter contact thermode). Interestingly, the signature response proved to be much more sensitive to variations in temperatures above versus below pain threshold, suggesting that it is partly specific to temperatures in the noxious range. The signature was then used to test the idea that psychological interventions aiming at reducing perceived pain exert their effects by decreasing the pattern of brain activity underlying “normal,” stimulus-driven, pain perception. More specifically, we examined the cerebral mechanisms underlying the effects of instructions to imagine thermal stimuli as more or less painful than they really are (Woo et al. 2015). Surprisingly, the NPS didn’t track the changes in perceived pain associated with these instructions, although it was highly sensitive to equivalent changes in pain that were driven by changes in temperature. By contrast with temperature, self-regulation of pain exerted most of its effects through the Nac (nucleus accumbens) and vmPFC. While these two structures are not typically part of the pain matrix, they are crucial for self-

related motivational processes (Roy et al. 2014), and would therefore perfectly fit Melzack's description of structures of the body-self neuromatrix. Interestingly, the connectivity between these two structures has been shown to be highly predictive of the development of chronic pain, suggesting that they may serve important cognitive and emotional aspects of pain that aren't bound to nociceptive stimulus intensity (see also [Chapter 10](#), this volume).

We also used the signature to test the hypothesis that physical and social pains share a common cerebral representation. As mentioned previously, this hypothesis was mainly based on observations that social rejection activates the same brain regions as those involved in the physical experience of pain. More specifically, a prior study conducted within our laboratory had shown that looking at pictures of an ex-romantic partner who had rejected one activated the thalamus, Ins, ACC, S2, and dpIns (dorso-posterior insula) (Kross et al. 2011). Based on the relative pain-specificity of the operculo-insular region (see above), we had initially concluded that these results indicated that rejection may literally share a common somatosensory representation with physical pain. However, when we re-examined these results with the NPS, we came to a different conclusion. Indeed, the NPS didn't respond to social rejection, even when we only considered the patterns of activity within the operculo-insular region (Wager et al. 2013). Therefore, despite the apparent similarity in the large-scale networks of structures activated by pain and social rejection, the two phenomena differ at the level of the fine-grain patterns of activity within each of these commonly activated structures. In Melzack's terms, this would mean that both social rejection and physical pain are represented within the same body-self neuromatrix, but that they have distinct neurosignatures.

To follow-up on this idea, we also developed a signature that could reliably distinguish looking at pictures of ex-partners versus friends (Woo et al. 2015). Indeed, one possible explanation for the NPS's lack of sensitivity to social rejection could have been that social rejection is simply more difficult to identify from patterns of brain activity. By developing a signature that was sensitive to rejection, but insensitive to physical pain (painful heat), we demonstrated that the two experiences were "separately modifiable," i.e., that they are represented by distinct patterns of brain activity. These findings also suggested that the NPS isn't sensitive to all sorts of salient stimuli (Iannetti and Mouraux 2010), such as pictures of ex-partners, but is rather specific to physical pain. Finally, in a strong test of the shared representation hypothesis, we systematically searched the brain for a potential "shared representation" between physical and social pain – areas in which the patterns that encode pain and social rejection are similar (Woo et al. 2015). We found evidence for shared representations in parts of the striatum,

temporoparietal junction, fusiform and parahippocampal gyrii, and posterior and retrosplenial cingulate cortices, suggesting the existence of shared processes. However, none of these shared processes were localized within any of the pain-matrix regions, suggesting that what is shared may be representations of negative affect and related mnemonic associations rather than activation of primary pain systems.

Finally, we examined the extent to which pain is represented similarly across individuals. Indeed, the NPS is amongst the first neural signatures capable of predicting a subjective experience in new individuals or new studies. One reason for this high inter-individual consistency in signature patterns may be that the seemingly dispersed nociceptive projections to the brain are also highly anatomically consistent across individuals (Dum et al. 2009). This again fits perfectly well with Melzack's idea of a partly genetically determined body-self neuromatrix. That being said, we should also expect important inter-individual differences in pain's representation due to gross inter-individual differences in brain anatomy and/or unique developmental histories. In order to test that hypothesis, we trained within-subject "idiographic" signatures, and tested the degree to which these idiographic signatures improved predictions made by the NPS across a set of seven different studies (Lindquist et al. 2015). Given an infinite amount of training data (e.g., hundreds of hours of imaging in the same subject), idiographic signatures should in principle always procure the optimal predictions. However, because we had limited training data (e.g., less than 1 hour of data per subject), predictions based solely on idiographic signatures were often imprecise. We therefore found that an optimized combination (weighted average) of idiographic signatures and of our group-based NPS always procured better predictions than idiographic signatures alone. Indeed, when idiographic signatures are found to be imprecise, a stronger weighting of the NPS can provide a fail-safe against poor predictions. Conversely, when idiographic signatures are found to be more precise, their contribution to the combined prediction can be increased so as to provide more flexibility to group-based NPS predictions. Altogether, these results support the idea that individual and contextual variations in the cerebral representation of pain exist, but that they are also constrained by the conserved neuroanatomical architecture of the nociceptive system.

A palette of signatures for a family of pains

Because pain subjectively feels like a very direct and unique experience, we are compelled to try to find an objective pattern of brain activity that would have a

perfect, one-on-one, correspondence with pain, i.e., to find “pain” in the brain. However, unlike objective phenomena that exist regardless of the presence of an observer to witness them (i.e., natural kinds), subjective phenomena like pain cannot exist without a subject to experience them (Barrett 2009; Searle 2007; see also Chapters 16, 18, 20, and 21 for more advanced discussions on this topic). This is well recognized by IASP’s definition of pain, which states that as long as an experience is described as pain, it should be recognized as pain (IASP 2014). In other terms, we are in pain as soon as we think that we are in pain. From that perspective, pain would therefore result from the act of categorizing an unpleasant, and potentially harmful, bodily sensation as an instance of “pain.” However, this doesn’t mean that nociceptive processes are irrelevant for understanding pain. Indeed, it is widely accepted that nociceptive sensations (e.g., burning, pricking, etc.) preferentially signal potential injury, like the alarm of a smoke detector may signal the presence of smoke. However, they do not, by themselves, constitute the experience of pain.

Indeed, pain rather seems to result from the combination of three core ingredients: (1) a bodily sensation, (2) negative affect, and (3) the appraisal – conscious or unconscious – that the experience indicates a potential injury. Therefore, various experiences that comprise these three basic ingredients in different proportions, or in slightly different kinds, could all be categorized as different instances of “pain”: the pain of a bee sting, the pain of a frostbite, the pain of a stomach ache, the pain of sore muscles, etc. However, as for most of our categories, it seems almost impossible to find a set of necessary and sufficient properties that perfectly define “pain” (see the note attached to the IASP definition of pain; IASP 2014). Rather, “pain” should be conceived as a collection of exemplars that share a “family resemblance” (Wittgenstein 2001/1953). It follows that some members of the pain “family,” like a stomach ache, may actually share more characteristics with closely related non-members of the family, like hunger, than with more distant members of the family, like the pain of a bee sting. In other words, it is possible that the category “pain” doesn’t perfectly carve nature at its joints.

If this is correct, then it would be virtually impossible to derive a signature that would be highly sensitive to all types of pain, and at the same time highly selective to pain. In order to reflect pain’s diversity, we could rather need a palette of signatures that would allow characterizing each type of pain with respect to other types of pain or pain-related phenomena, such as thermal pain, hunger, social rejection, etc. We have just begun to lay some of the methodological foundations for this vast program of research, which was initially envisioned by Melzack almost twenty years ago. However, it was just

with the development of multivariate pattern analyses that we acquired the ability to decode the cerebral representation of pain, and to begin addressing Melzack's initial question: how can the brain generate the experience of pain in the absence of nociception (Woo et al., submitted)? As intuited by Melzack, the answer to that question may force us to imagine another, more representational layer of neuroscientific explanation, measurable in terms of signature patterns of distributed brain activity. Although it is too soon to say where this paradigm shift is taking us, we hope that it will enlighten our comprehension of the role of the brain in normal and pathological chronic pain syndromes, and ultimately enhance assessment and treatment.

Related topics

[Chapter 6](#): Advances in the neuroscience of pain (Apkarian)

[Chapter 9](#): A view of pain based on sensation, meaning, and emotions (Price)

[Chapter 10](#): Pathophysiological mechanisms of chronic pain (Thacker and Moseley)

[Chapter 16](#): Robot pain (Mandik)

[Chapter 18](#): Pain: perception or introspection? (Aydede)

[Chapter 20](#): Pain and incorrigibility (Langland-Hassan)

[Chapter 21](#): Can I see your pain? An evaluative model of pain perception (de Vignemont)

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